

Control work 7, and work on the mistakes

Probability and distributions in one paper, then the block drawn as a decision tree.

LESSON 90–91

2 × 40 MIN

ALGEBRA 11, NAZORAT ISHI 7

P&S 1 REVIEW

LESSON CLOCK

3

45'

12'

20'

5'

Instructions	3 min
The paper	45 min
Answers	12 min
Diagnosis and rewrite	20 min
The decision tree	5 min

LEARNING OBJECTIVES

- Apply the counting, addition and multiplication rules under time.
- Compute a conditional probability with the right denominator.
- Use the binomial formula and the normal 68–95–99.7 rule.
- Name each lost mark and rewrite the solution in full.

TEXTBOOK

National	pp. 407–410
Cambridge	P&S 1, pp. 151–154
Workbook	Control paper A7

01

Explanation

The paper — 40 marks, 45 minutes

Q	TASK	MARKS	FROM
1	A committee of 4 from 6 men and 5 women: $P(\text{exactly 2 women})$ and $P(\text{at least 1 woman})$	7	L79–80
2	Given $P(A) = 0.5$, $P(B) = 0.4$, $P(A \cup B) = 0.7$: find $P(A \cap B)$, $P(A \cap B')$ and say whether the events are independent	6	L81–82
3	A bag of 7 red and 5 blue, two drawn without replacement: the tree, $P(\text{same colour})$, $P(\text{second red})$	7	L83–85
4	Two machines, 65/35\% of output with 2/6\% faulty: $P(\text{faulty})$ and $P(\text{machine B} \mid \text{faulty})$	7	L83–85
5	$X \sim B(12, 0.25)$: find μ , σ , $P(X = 3)$ and $P(X \geq 1)$	7	L86–89
6	Masses are $N(500, 15^2)$ g: the proportion above 530 g, and between 485 and 515 g	6	L86–89

WHERE THE MARKS ACTUALLY GO

Q1 carries one mark for the complement rather than four cases; Q3 one for changing the second fork; Q4 one for the denominator $P(\text{faulty})$ rather than 1; Q5 one for the complement in $P(X \geq 1)$. Four of the forty marks are for one decision each.

The decision tree for the whole block

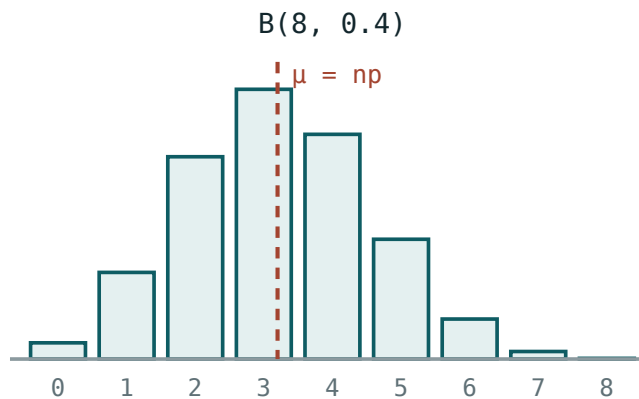
THE QUESTION SAYS	USE	FORMULA
“committee”, “hand”, “selection”	combinations	$\frac{C(a, j) \cdot C(b, k)}{C(a+b, j+k)}$
“or”, “either”	the addition rule	$P(A) + P(B) - P(A \cap B)$
“and”, “then”	the multiplication rule	$P(A)P(B A)$
“at least one”	the complement	$1 - P(\text{none})$
“given that”	conditional	$\frac{P(A \cap B)}{P(B)}$
“exactly k out of n ”	binomial	$C(n, k)p^k(1-p)^{n-k}$
“a measured quantity”, “heights”, “masses”	normal	$z = \frac{x - \mu}{\sigma}$

THE FIVE SLIPS THAT ACCOUNT FOR MOST LOST MARKS

- counting m and n by different methods;
- adding without subtracting the overlap;
- keeping the denominator fixed in a without-replacement tree;
- swapping $P(A | B)$ and $P(B | A)$;
- treating a without-replacement count as binomial.

Looking forward

Lessons 92–100 are the Cambridge revision block: complex numbers and the Argand diagram, loci in the complex plane, and differential equations. It is the last new mathematics of the school course, and the first mathematics of the university one.



$$P(X = k) = C(n, k) p^k (1 - p)^{n-k}$$

The last picture of the probability block — and the shape the normal curve takes over.

ONE HABIT TO CARRY FORWARD

Probability rewarded naming the situation before choosing a formula. Complex numbers and differential equations reward naming the **form** before choosing a method — Cartesian or polar, separable or not. The habit is the same; only the vocabulary changes.

02 Worked examples

EXAMPLE 1 Model answer, Q1: a committee of 4 from 6 men and 5 women — $P(\text{at least 1 woman})$.

① $n = C(11, 4) = 330$

② $P(\text{no woman}) = \frac{C(6, 4)}{330} = \frac{15}{330}$

The complement.

③ $P = 1 - \frac{15}{330} = \frac{315}{330} = \frac{21}{22}$

ANSWER

$$\frac{21}{22} \approx 0.955$$

EXAMPLE 2 Model answer, Q4: machines 65\%/2\% and 35\%/6\%.

$$① \quad P(F) = 0.65(0.02) + 0.35(0.06)$$

$$② \quad = 0.013 + 0.021 = 0.034$$

$$③ \quad P(B | F) = \frac{0.021}{0.034}$$

$$④ \quad \approx 0.618$$

Machine B makes a third of the output but most of the faults.

ANSWER

0.034 and ≈ 0.618

EXAMPLE 3 Model answer, Q5: $X \sim B(12, 0.25)$.

$$① \quad \mu = 3$$

$$② \quad \sigma^2 = 12(0.25)(0.75) = 2.25 \Rightarrow \sigma = 1.5$$

$$③ \quad P(X = 3) = C(12, 3)(0.25)^3(0.75)^9 \approx 0.258$$

$$④ \quad P(X \geq 1) = 1 - 0.75^{12} \approx 0.968$$

Complement.

ANSWER

$\mu = 3, \sigma = 1.5, 0.258, 0.968$

03 Interactive model

Work Q4 twice — once dividing by 0.034, once by 1 — and let the class say which question each answers.

Twelve questions on the probability block

A committee of 3 from 10:

30

120

720

1000

Order matters means:

$C(n, k)$

$P(n, k)$

$n!$

$k!$

$P(A \cup B)$:

$P(A) + P(B)$

$P(A) + P(B) - P(A \cap B)$

$P(A)P(B)$

$1 - P(A)$

$(A \cup B)'$:

$$A' \cup B'$$

$$A' \cap B'$$

$$A \cap B$$

$$U$$

$P(A \cap B)$ in general:

$$P(A)P(B)$$

$$P(A)P(B | A)$$

$$P(A) + P(B)$$

$$0$$

Without replacement, the denominator:

stays

drops by one

doubles

is 1

Total probability sums over:

any events

a partition

independent events

nothing

$P(B | A)$ divides by:

$P(B)$

$P(A)$

1

$P(A \cup B)$

$B(n, p)$ needs p :

small

constant

large

variable

Its mean:

np

$np(1 - p)$

p

$\sqrt{np}$

Within two σ lies:

68\%

95\%

99.7\%

50\%

$z = 2$ means:

twice the mean

two σ above the mean

a probability of 2

nothing

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Combination	Kombinatsiya	Сочетание
Conditional probability	Shartli ehtimollik	Условная вероятность
Total probability	To'la ehtimollik	Полная вероятность
Binomial distribution	Binomial taqsimot	Биномиальное распределение
Standard deviation	Standart chetlanish	Стандартное отклонение
z-score	z-baho	z-оценка
Target	Maqsad	Цель

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 is quickest by:

four cases

the complement

a tree

a table

Q3 needs the second fork to:

stay the same

change

be doubled

be ignored

Q4's denominator is:

1

$P(\text{faulty})$

0.35

0.06

Lessons 92–100 revise:

probability

complex numbers and differential equations

trigonometry

integration

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $C(11, 4)$

ANSWER 330

2. $C(6, 4)$

ANSWER 15

3. $P(A \cap B)$ when $P(A) = 0.5$, $P(B) = 0.4$, $P(A \cup B) = 0.7$

ANSWER 0.2

4. Are those events independent?

ANSWER Yes — $0.5 \times 0.4 = 0.2$

5. μ of $B(12, 0.25)$

ANSWER 3

6. σ of $B(12, 0.25)$

ANSWER 1.5

7. z for 530 in $N(500, 15^2)$

ANSWER 2

Medium · the standard question

7 PROBLEMS

1. $P(\text{exactly 2 women})$ on a committee of 4 from 6 men and 5 women

ANSWER $\frac{150}{330} = \frac{5}{11}$

2. $P(\text{at least 1 woman})$ on the same committee

ANSWER $\frac{21}{22}$

3. Bag of 7 red, 5 blue: $P(\text{same colour})$ without replacement

ANSWER $\frac{31}{66}$

4. Same bag: $P(\text{second red})$

ANSWER $\frac{7}{12}$

5. Machines $65\frac{1}{2}\%$ and $35\frac{5}{6}\%$: $P(\text{faulty})$

ANSWER 0.034

6. Same: $P(B \mid \text{faulty})$

ANSWER ≈ 0.618

7. $P(X = 3)$ for $B(12, 0.25)$

ANSWER ≈ 0.258

Hard · stretch and reason

7 PROBLEMS

1. $P(X \geq 1)$ for $B(12, 0.25)$

ANSWER ≈ 0.968

2. Proportion of $N(500, 15^2)$ above 530

ANSWER $\approx 2.3\%$

3. Proportion of $N(500, 15^2)$ between 485 and 515

ANSWER $\approx 68\%$

4. A committee of 5 from 6 men and 5 women with more women than men: P

ANSWER $\frac{181}{462} \approx 0.392$

5. Three components fail with probability 0.05 each: P at least one fails

ANSWER ≈ 0.143

6. For $B(n, 0.25)$ with $\sigma = 3$, find n

ANSWER $n = 48$

7. A test is 98% accurate; 0.5% have the disease: $P(\text{disease} \mid \text{positive})$

ANSWER ≈ 0.198

07 Homework

Homework — 4 tasks

Bring the decision tree to Lesson 92; complex numbers start from a blank page.

1. Rewrite in full every control-work question that lost a mark, naming the slip in the margin.
2. Copy the decision-tree table and add one worked example of your own to each row.
3. A bag holds 8 white and 4 black balls; three are drawn without replacement. Find the probability that all three are white and the probability that at least one is black.
4. Write your target for the revision block in one checkable sentence, and date it.